Equipment Failure Prediction

Logistic Regression — Project Report

1. Problem Statement

Industrial equipment failure leads to unplanned downtime and high repair costs. Early warning signs are often present in operational sensor data before a failure occurs. This project builds a binary classification model using Logistic Regression to predict whether a machine will fail within its next operating window, based on real-time sensor and maintenance data.

2. Dataset Description

The dataset (*dataset_06_equipment_failure_warning.csv*) contains 1,000 records with 6 input features and one binary target label.

Feature	Description	Type
temperature	Operating temperature reading	float
vibration	Vibration sensor reading	float
pressure	Pressure sensor reading	float
runtime_hours	Continuous runtime since last cycle start	float
maintenance_gap_days	Days since last maintenance	int
power_draw	Power consumption reading	float
target	0 = No Failure, 1 = Failure (label)	int

3. Exploratory Data Analysis

The dataset was inspected for shape, data types, missing values, duplicate rows, and class balance before any modeling was performed.

Findings:

- No missing values were found across any of the 7 columns.
- No duplicate rows were present in the dataset.
- The target variable is well-balanced: approximately 500 'No Failure' and 500 'Failure' records (50/50 split), which means accuracy alone is a meaningful metric here and no class-imbalance correction (e.g. SMOTE, class weighting) was required.
- Feature ranges vary considerably (e.g. power_draw spans ~20 to ~1940), confirming the need for feature scaling before training a distance/gradient-sensitive model like Logistic Regression.

4. Preprocessing

- **Train/test split performed first** (80/20, stratified on target, random_state=42) — this order matters because fitting the scaler on the full dataset before splitting would leak information about the test set's distribution into training, artificially inflating performance.
- **Feature scaling** using StandardScaler, fit only on the training set and then applied to transform both train and test sets.
- No categorical encoding was required, as all features are numeric.

5. Model Configuration

A Logistic Regression classifier (scikit-learn) was trained with the following configuration:

- Algorithm: LogisticRegression
- max_iter: 1000
- random_state: 42
- Solver: default (lbfgs)

6. Results & Evaluation

Metric	Value
Accuracy	71.5%
Recall (Failure class)	66.0%
ROC-AUC	0.78

Confusion Matrix

	Predicted: No Failure	Predicted: Failure
Actual: No Failure	77 (TN)	23 (FP)
Actual: Failure	34 (FN)	66 (TP)

Out of 200 test samples, the model correctly identified 66 of 100 actual failures (recall = 66%), while missing 34 (false negatives). It correctly identified 77 of 100 non-failure cases, with 23 false alarms. The ROC-AUC of 0.78 indicates the model has reasonably good separability between the two classes, though there is clear room for improvement.

7. Coefficient Interpretation

Feature	Coefficient	Direction of Influence
pressure	+0.550	Higher pressure → higher failure risk
temperature	+0.546	Higher temperature → higher failure risk
power_draw	-0.459	Higher power draw → lower failure risk
maintenance_gap_days	+0.442	Longer gap since maintenance → higher failure risk
runtime_hours	-0.415	Higher runtime → lower failure risk
vibration	-0.365	Higher vibration → lower failure risk

Pressure and temperature are the strongest positive predictors of failure, meaning machines running hotter and under higher pressure are flagged as higher risk. Interestingly, vibration and runtime_hours show a negative relationship with failure in this dataset — this may reflect that well-maintained, actively-used machines vibrate within a normal operating band, whereas the model picks up failure risk more strongly from thermal and pressure stress than from vibration alone.

8. Limitations & Future Work

- Recall of 66% means roughly 1 in 3 actual failures is missed — in a real predictive-maintenance deployment, false negatives are more costly than false alarms, so this model's threshold or the algorithm choice itself would need further tuning before production use.

• Logistic Regression assumes a linear decision boundary between classes; non-linear models such as Random Forest or Gradient Boosting could potentially capture more complex interactions between sensor readings.

• No hyperparameter tuning (e.g. regularization strength C, class weighting) was performed in this baseline version.

• The model was trained on a single static dataset snapshot; in a real deployment, sensor drift over time would require periodic retraining.

9. Conclusion

This project demonstrates a complete, leakage-safe logistic regression pipeline for predicting equipment failure from sensor data, achieving 71.5% accuracy and a ROC-AUC of 0.78. Pressure and temperature emerged as the strongest indicators of failure risk. While the model shows meaningful predictive signal, its recall indicates it would need further refinement — through better feature engineering, alternative algorithms, or threshold adjustment — before being reliable enough for real-world predictive maintenance decisions.